WATER PURIFICATION



R26-IT-005

Progress Presentation 1

SLIIT
FACULTY OF COMPUTING

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Comments for the Proposal and how you/team addressed them

STUDENT'S NAME & ID	COMMENTS FOR THE PROPOSAL	HOW YOU/TEAM ADDRESSED THEM
R.M.N.A. RATHNAYAKE IT22193940	• Propose to test the quality of the water using sensor data from the water collected using water sources • The member 2's component can be incoperated with the same component	• The component scope was refined to focus on intelligent water purification and monitoring.The system was designed to automatically adjust mineral composition based on sensor feedback. IoT-enabled monitoring ensures real-time control of pH levels and mineral concentration, enabling consistent alkaline water output.
W.N.N. DE SILVA IT22211750	• Propose to classify water types using sensors • Scope needs to be defined properly, could be incopeated with Member 1's component	• A machine learning model was implemented to classify water types using sensor data such as pH, TDS, turbidity, conductivity, and temperature. Training and testing were done using collected datasets to improve classification accuracy for different water categories (safe, unsafe, distilled, alkaline, etc.)
D. DODANWELA IT22212122	• Propose to develop an IoT device • Check with portability of the device	• The system design was refined to ensure the IoT device is compact, portable, and suitable for household use cases. The device integrates a water purification and boiling mechanism that produces clean, safe drinking water.
W.A.S.P.D. WIJESINGHA IT22295460	• Check with the feasibility of the component • Propose to develop an IoT device	• The system incorporates an atmospheric water generation mechanism, where moisture from air is collected and condensed into liquid water, making the overall solution practical for real-world implementation and sustainable water production.

WATER PURIFICATION



PROBLEM

- Traditional water purification and monitoring methods rely on manual testing, fixed purification settings, and separate monitoring systems. Existing systems cannot intelligently identify different water types or automatically optimize purification and mineral balancing based on real-time sensor data.

IMPACT

- Develop an ai and iot-based smart water management system capable of automatically identifying water types using multi-sensor data, optimizing purification processes, monitoring water quality in real time, and supporting intelligent mineral balancing and alkaline water adjustment.

DESIGN EXCELLENCE

- The proposed system integrates IoT sensors, machine learning, automated purification, alkaline optimization, and real-time monitoring into a unified smart water ecosystem.
- The system improves water quality analysis, reduces manual intervention, enhances purification efficiency, and provides intelligent decision-making based on live sensor readings.

IT22295460 | W.A.S.P.D. WIJESINGHA



ATMOSPHERIC WATER GENERATION (IoT-BASED)

Specialization : Information Technology



How the Solution Addresses the Sub-Problem

- Solves clean water scarcity using moisture from air
- IoT-based smart atmospheric water generator
- Optimizes cooling and condensation cycles
- Supports low-power and energy-efficient operation
- Enables automated water generation and collection
- Integrates with purification and AI-based water testing modules

User Requirements Addressed by the Solution

- Can be use to provide clean water in remote or dry areas
- Enables real-time monitoring through IoT Supports:
  - Automatic system control
  - Smart energy optimization
  - Reduces dependency on traditional water sources
  - Supports sustainable and smart water generation
  - Easily integrates with the smart water ecosystem

Design Excellence / Contribution



- Smart atmospheric water generation approach: IoT-based environmental monitoring, temperature monitoring, humidity detection, and water level measurement.
- Energy-efficient condensation and cooling control with modular system design for easy integration.
- Supports integration with water purification modules, AI-based water quality analysis, and smart monitoring dashboards with real-time monitoring and remote access capability.

User Feedback on Prototype

- Users found the system innovative and useful for water-scarce areas
- Real-time IoT monitoring improved user confidence and usability
- Users appreciated the automatic water generation and collection process

IT22212122 | D. DODANWELA



HOUSEHOLD DISTILLED WATER MAKER

Specialization : Information Technology

How the Solution Addresses the Sub-Problem



- Solves clean water scarcity by converting unclean water into purified distilled water through a controlled boiling and condensation process.
- IoT-based smart household distilled water generation system optimizes heating, boiling, and condensation cycles for improved efficiency.
- Supports low-power and energy-efficient operation through smart control mechanisms.

User Requirements Addressed by the Solution

- Provides clean and safe drinking water at household level by purifying unclean water.
- Enables real-time monitoring through IoT, supporting automatic system control smart energy optimization, and improved operational efficiency.
- Reduces dependency on external or traditional water sources.
- Supports sustainable and smart water generation within households.
- Easily integrates with the smart water ecosystem for future expansion and upgrades.



Design Excellence / Contribution

- To design, develop, and validate a smart, energy-optimized household distilled water generation system that converts unclean water into purified distilled water through a controlled boiling and condensation process.

- The system integrates IoT-based monitoring to enhance user convenience and operational efficiency.

User Feedback on Prototype

- Users found the smart distillation concept useful for household water purification
- Users liked the ability to control and monitor the system remotely
- Positive feedback received for integrating smart automation into traditional distillation methods
- Suggested improvements:
  - Reduce distillation time further
  - Add water quality indicators and maintenance alerts

IT22193940 | R.M.N.A.RATHNAYAKE



ALKALINE WATER GENERATION USING IoT-BASED MINERAL BALANCING

Specialization : Information Technology

How the Solution Addresses the Sub-Problem



- Addresses the need for safe and healthy drinking water by converting regular water into alkaline water through IoT-controlled mineral balancing.
- The system optimizes pH adjustment and mineral infusion cycles to maintain consistent water quality.
- Enables fully automated alkaline water generation with minimal manual intervention.
- Integrates with water quality monitoring and AI-based analysis modules for improved safety and consistency.

User Requirements Addressed by the Solution

- Provides automated alkaline water generation using smart mineral balancing
- Reduces manual intervention in alkaline water preparation
- Ensures consistent water quality and balanced mineral content
- Enables IoT-based monitoring and system control
- Supports customized water optimization for different user needs
- Integrates with AI/ML modules for intelligent decision-making
- Improves efficiency and reliability of alkaline water production

Design Excellence / Contribution



- To design, develop, and validate a smart IoT-based alkaline water generation system using mineral balancing techniques to produce healthy drinking water with controlled pH levels.

- The system integrates IoT-based monitoring and mobile application control to enhance user convenience and operational efficiency while ensuring consistent water quality through automated mineral adjustment.

User Feedback on Prototype

- Users found the automated alkaline water adjustment feature useful and innovative
- Users appreciated the smart mineral balancing and reduced manual effort
- The IoT-based monitoring interface was easy to understand and use
- Users liked the idea of customized alkaline water generation
- Automatic adjustment features were considered more efficient than manual methods
- Suggested improvements:
  - Improve accuracy of mineral dosing mechanisms
  - Add personalized alkaline level recommendations

IT22211750 | DE SILVA W.N.N.



DIFFERENT TYPES OF WATER IDENTIFIER – ML & AI

Specialization : Information Technology

How the Solution Addresses the Sub-Problem



- The solution addresses the difficulty in identifying different types of water by introducing an AI/ML-based classification system that analyzes real-time sensor data.
- Using inputs such as pH, TDS, turbidity, conductivity, and temperature, the model classifies water into categories (e.g., safe drinking, alkaline, distilled, contaminated).
- This removes manual testing and enables fast, automated, and accurate water quality identification at each purification stage.

User Requirements Addressed by the Solution

- Real-time identification of water type using sensor data
- Accurate classification of safe and unsafe water
- Simple and automated decision-making without manual testing
- Continuous monitoring during purification process
- Clear output indicators for user understanding

Design Excellence / Contribution



- The design integrates machine learning classification models with IoT sensor data pipelines, enabling intelligent decision-making.
- Feature selection from multiple water quality parameters improves prediction accuracy.
- The system is modular, allowing easy integration with other subsystems such as purification, distillation, and atmospheric water generation.

User Feedback on Prototype

- Users reported that the system was easy to understand and provided clear water type results in real time
- The automated classification reduced the need for manual water testing
- The system increased user trust in water safety decisions
- Visual indicators and simple labels improved usability and clarity of results
- Suggested Improvements:
  - Expand support for additional water types and edge cases
  - Increase accuracy of the ML model with larger and more diverse datasets



Commercialization & Sustainability

- Can be marketed as a smart IoT-based water purification system
- Suitable for households, rural communities, and small institutions
- Low-power operation ensures energy-efficient and sustainable usage
- Modular system allows easy scaling and product expansion

Customer Persona & Market

- Households needing safe and clean drinking water
- Rural and semi-urban communities with limited water infrastructure
- Schools, clinics, and small institutions requiring purified water
- Smart home users interested in IoT-based appliances

Uniqueness & Cost Recovery

- Integrates IoT monitoring with water purification process
- Real-time water quality tracking via mobile application
- Automated purification reduces manual intervention and maintenance cost
- Cost recovery through device sales, maintenance, and service subscriptions

THANK YOU

ANY QUESTIONS ?